Module 1: Advanced Linear Algebra Topics

1. Invertibility of Matrices (Singular vs. Non-singular)

- A square matrix A is non-singular (invertible) if there exists A■^{1} such that AA■^{1} = I.
- Determinant Condition: det(A) ≠ 0.
- Full Rank Condition: rank(A) = n for an n×n matrix.
- Linearly Independent Columns: Columns form a basis for R■.
- Singular Matrix: det(A) = 0.
- If singular, the system Ax = 0 has non-trivial solutions.
- Non-singular matrix implies unique solution for Ax = b.

2. Rank–Nullity Theorem

- Statement: For an m×n matrix A, Rank(A) + Nullity(A) = n.
- Rank: Dimension of the Image (column space).
- Nullity: Dimension of the Kernel (null space).
- Every dimension in input space either contributes to rank or nullity.

3. Basis and Dimension (Higher Reasoning)

- Basis: Set of vectors that are linearly independent and span the vector space.
- Dimension: Number of vectors in any basis of V.
- If dim(V) = n, any set of n independent vectors forms a basis.
- Maximum independent vectors in R■ is n.
- Standard basis vectors in R^3 are orthogonal and unit length.

4. Kernel and Image (Functional Perspective)

- Kernel: Set of vectors x such that T(x) = 0.
- Image: Set of all possible outputs of T.
- Dimension of Image equals rank of transformation matrix.

- System Ax = b is consistent only if b lies in the Image of A.

5. Singular Matrices and System Behavior

- If A is non-singular, Ax = 0 has only trivial solution.
- If A is singular, Ax = 0 has infinitely many non-trivial solutions.
- Diagonalization: $A = PDP^{-1}$ when eigenvectors form a basis.
- If A is singular, at least one eigenvalue is zero.